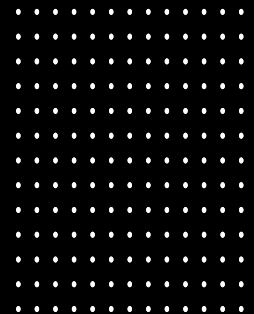
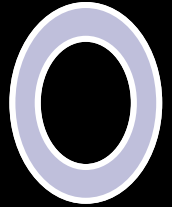
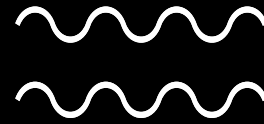


The Endocrine System Lecture 5

E. Reproductive Hormones

- **General concepts**
- Reproductive hormones in the male
- Reproductive hormones in the female



Human Sex Characteristics

- Primary sex characteristics:

Directly related to reproduction and include the genitalia.

- Secondary sex characteristics:

Other attributes that generally distinguish one sex from the other, such as: *voice quality; abundance of facial hair; breast development.*

Overview of Reproductive Organs (Primary Sex Characteristics)

Two basic types of reproductive system organs –

1. **Gonads = primary sex organs:** ovary and testis

a. Produce the **gametes**.

b. Produce the **sex hormones** that –

1. *regulate the reproductive process.*

2. *regulate development and function of the other reproductive organs.*

2. **Secondary sex organs;** such as the uterus and the penis.

a. transport, b. protection, and c. nourishment of the gametes, embryo or young.

Overview of Sex & Reproductive Hormones

1) Steroid hormones produced from cholesterol

-Biosynthetic Pathway: Cholesterol -> progesterone -> androgens -> estrogens

- ❖ Androgens - mainly testosterone. More important in males.
- ❖ Estrogens - mainly 17-beta-estradiol. More important in females.
- ❖ Progesterone - more important in females.

2) 2 peptide hormones produced in the pituitary

- oxytocin and prolactin; function in childbirth and lactation.

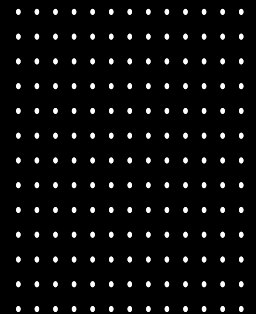
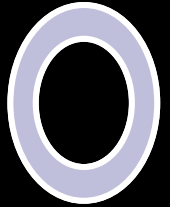
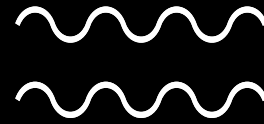
Table 9-9 Important mammalian reproductive hormones

Hormone	Tissue of origin	Structure	Target tissue	Primary action	Regulation
Primary sex hormones					
Estradiol-17 β (estrogens)	Ovarian follicle, corpus luteum, adrenal cortex	Steroid	Most tissues	Promotes development and maintenance of female characteristics and behavior, oocyte maturation, and uterine proliferation	Increased FSH and LH levels stimulate secretion
Progesterone	Corpus luteum, adrenal cortex	Steroid	Uterus, mammary glands	Maintains uterine secretion; stimulates mammary duct formation	Increased LH and prolactin levels stimulate secretion
Testosterone (androgens)	Testes (Leydig cells), adrenal cortex	Steroid	Most tissues	Promotes development and maintenance of male characteristics and behavior and spermatogenesis	Increased LH level stimulates secretion
Other Hormones					
Oxytocin	Posterior pituitary	Nonapeptide	Uterus, mammary glands	Promotes smooth muscle contraction and milk ejection	Cervical distention and suckling stimulate release; high progesterone inhibits release
Prolactin (PL)	Anterior pituitary	Peptide	Mammary glands (alveolar cells)	Increases synthesis of milk proteins and growth of mammary glands; elicits maternal behavior	Continuous secretion of PL-inhibiting hormone (PIH) normally blocks release; increased estrogen and decreased PIH secretion permit release

The Endocrine System Lecture 5

E. Reproductive Hormones

- General concepts
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- Reproductive hormones in the female



The Sex Organs of the Male

They are external and have two major parts:

1) *Penis*

2) *Scrotum (why is it outside?):*

a. *Two testes* – spermatogenesis occurs in seminiferous tubules; Leydig cells make testosterone.

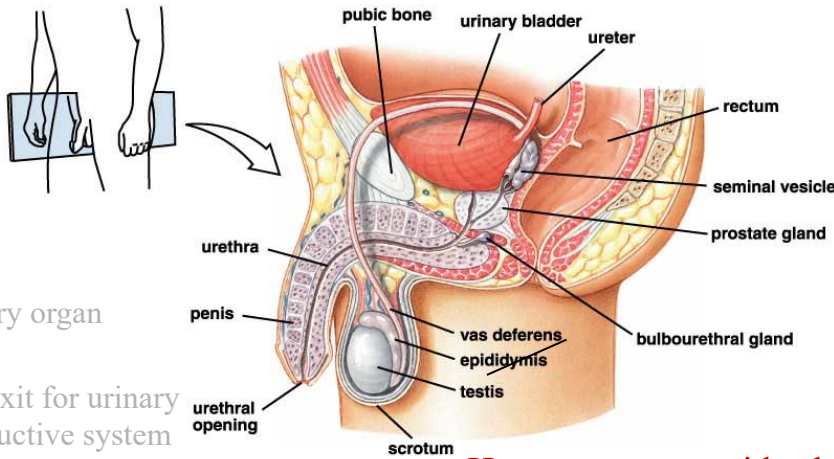
b. *Epididymis*, site of final sperm maturation, which takes several weeks.

c. *Vas deferens*, sperm transport. Sperm– through inguinal canal to abdominal cavity out urethra; fluids added along the way.

Seminiferous: seed producing

Semin- = seed

-ferous = producing



Copulatory organ

Common exit for urinary and reproductive system

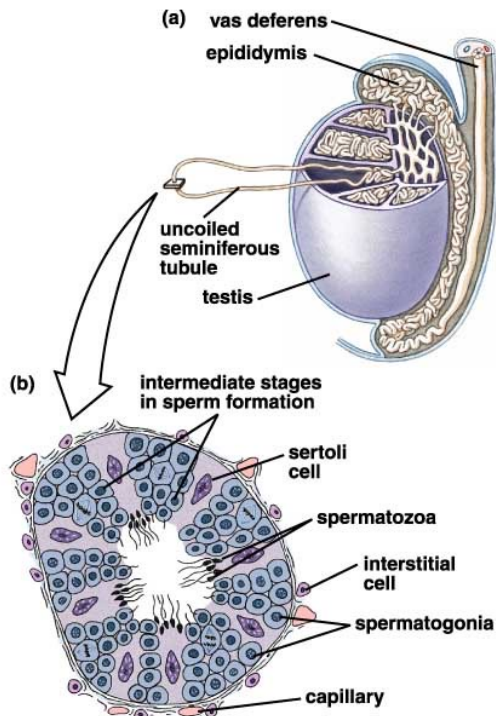
Add 60% of fluid to semen; contains fructose (energy source), prostaglandins, proteins to coagulate semen in vagina

Adds seminal fluids neutralize acidic urethra and vagina

Houses testes outside abdominal cavity since sperm production does not occur at normal body temperature

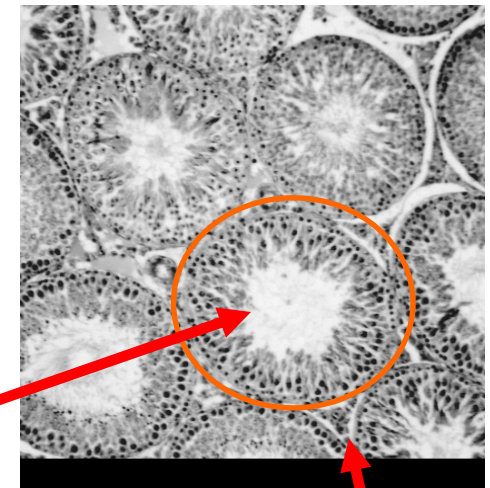
Male primary sex organ & associated organs

The Leydig Cell



- Interstitial cells
- production of testosterone
- respond to LH

Seminiferous tubules



Leydig cells

Spermatogenesis

- The process of sperm production. Germ cells undergo a series of divisions and differentiations to become mature sperm.
- It takes place in the seminiferous tubules. Seminiferous tubules are made up of
 - 1) Germ cells - making sperms
 - 2) Sertoli cells - supporting cells
- The average college-age males produce about 60 million sperms/ml semen. If sperm counts < 35 million/ml, fertility is markedly impaired.

Function of Sertoli cells:

1. Form blood-testes barrier.
2. Production of seminiferous tubular fluid, which stimulates and nourishes developing germ cells.
3. Synthesis of inhibin.

Sperm counts are falling (due to environmental pollutants) ?

Temporal trends in sperm count: a systematic review and meta-regression analysis

Hagai Levine ✉, Niels Jørgensen, Anderson Martino-Andrade, Jaime Mendiola, Dan Weksler-Derri, Irina Mindlis, Rachel Pinotti, Shanna H Swan

Human Reproduction Update, Volume 23, Issue 6, 1 November 2017, Pages 646–659,

<https://doi.org/10.1093/humupd/dmx022>

Published: 25 July 2017 **Article history** ▼

YES



NATURE REVIEWS | **UROLOGY**

Declining sperm counts — the never-ending story

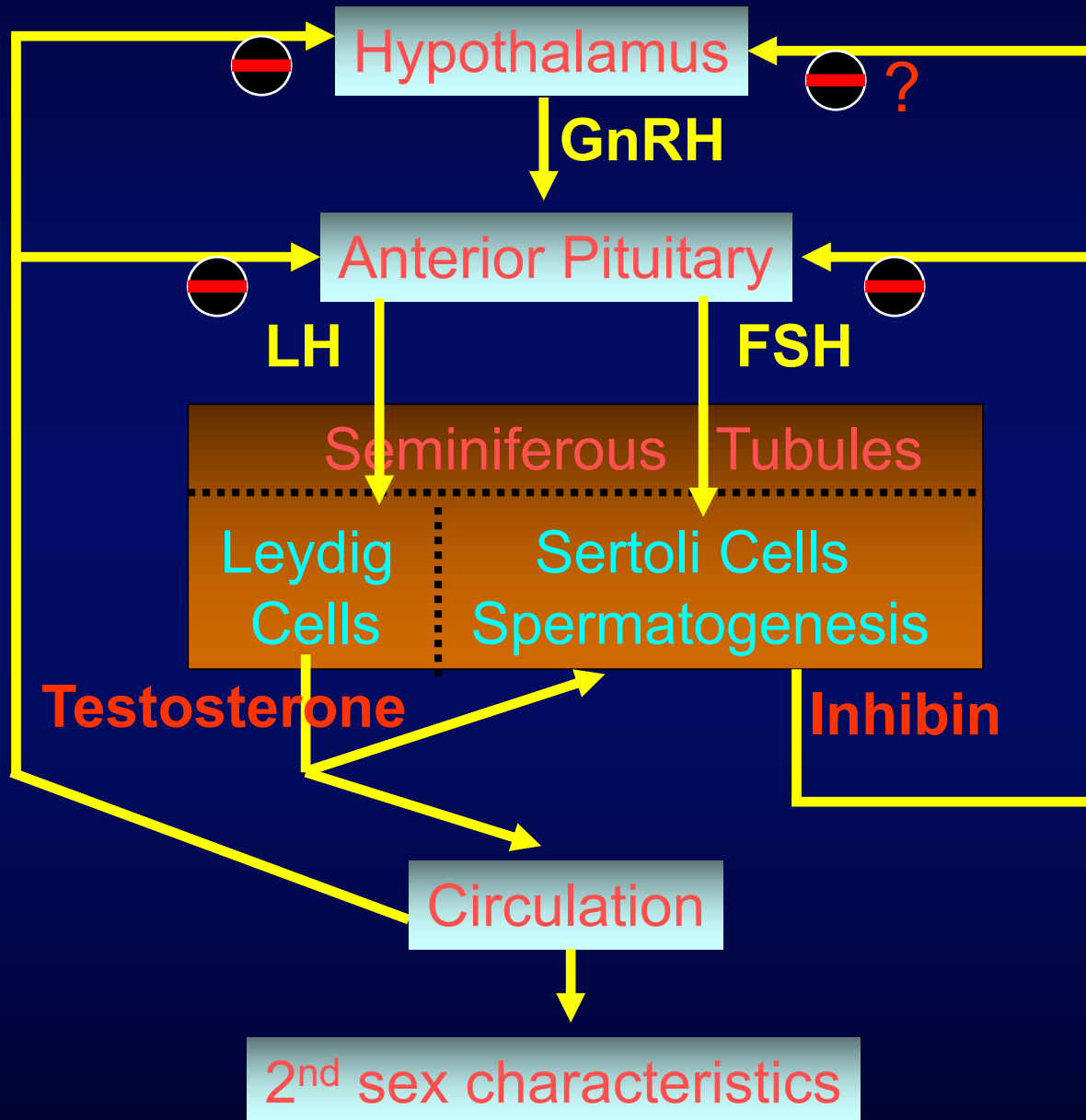
Jens Peter Bonde and Egbert te Velde

A recent systematic review on worldwide declining trends in sperm counts has fuelled alarming reports in national and international news media. However, methodological issues exist with data gathering and analysis precluding any conclusion and no solid data exist to indicate increasing frequency of couple infertility during past decades.

Refers to Levine, H. et al. Temporal trends in sperm count: a systematic review and meta-regression analysis. *Hum. Reprod. Update* <http://dx.doi.org/10.1093/humupd/dmx022> (2017)

NO

Neuroendocrine Control of Spermatogenesis



Both LH and FSH are required for spermatogenesis.

LH stimulates testosterone synthesis in Leydig cells. Testosterone has inhibitory feedback effect on the release of GnRH, LH, & FSH.

FSH binds receptors on Sertoli cells and stimulates the spermatogenesis in a process that also requires testosterone. Inhibin, a peptide hormone from Sertoli cells, has inhibitory feedback effect on the release of FSH.

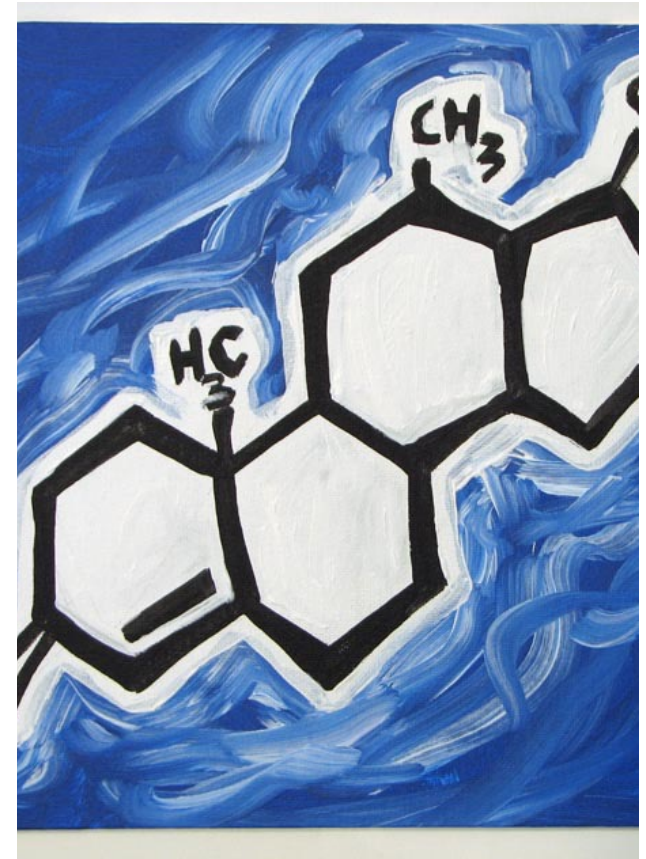
Control of GnRH by Sensory Stimuli

- Visual, olfactory, or tactile stimuli
- Thoughts and moods
- Stress, e.g. protracted aerobic exercise, inhibits GnRH secretion in both males and females=>inhibition of sperm production in the male and disruption of menstruation in the female.

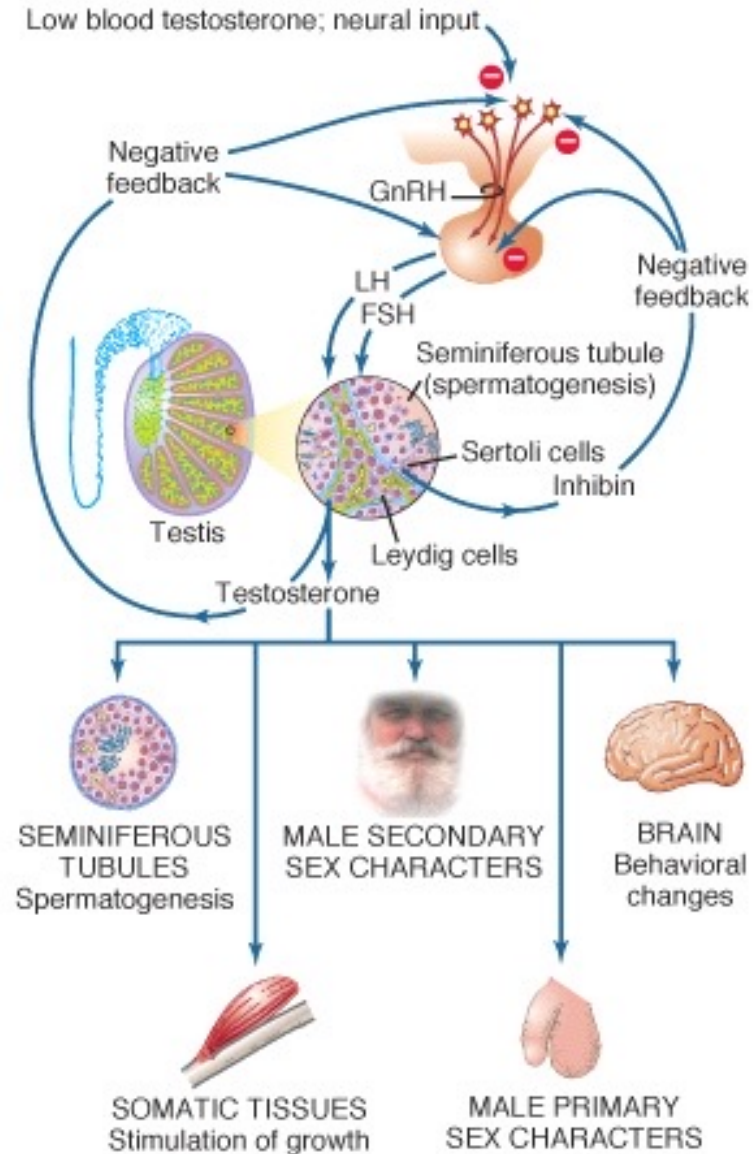
Actions of Testosterone

- The primary androgen produced in testes (Leydig cells) with following actions:

- 1) In prenatal development, it triggers development of primary male sex characteristics.
- 2) At puberty, enlargement of male sex organ (penis, scrotum, & testes) and development of secondary sex characteristics.
- 3) Spermatogenesis.
- 4) General growth and protein synthesis - especially muscle proteins, giving greater muscle mass in males.
- 5) Behavioral changes - Aggression.



Physiology of Testosterone (Summary)



Quizzes

Which of the following sentences is TRUE:

- 1) All reproductive hormones are steroids.
- 2) All steroids are reproductive hormones.
- 3) All reproductive hormones are secreted from sex organs.
- 4) Androgens are present only in males.
- 5) None of the above.

Which of the following concerning Leydig cells is FALSE:

- 1) secrete androgens.
- 2) are the target of LH.
- 3) are outside of the seminiferous tubules.
- 4) secrete inhibin.

Quizzes

Sertoli cells perform all of the following functions except:

- 1) binding LH.
- 2) contributing to seminiferous tubule fluid production.
- 3) nourishing developing sperm.
- 4) secreting inhibin.
- 5) forming blood-testes barrier.

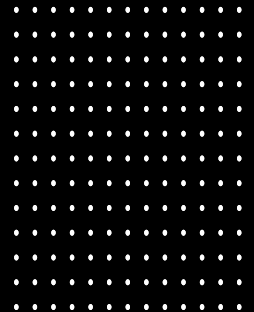
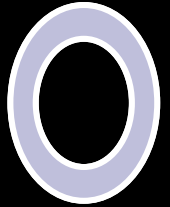
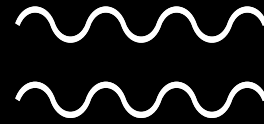
Testosterone:

- 1) is secreted only by Sertoli cells.
- 2) is secreted only by Leydig cells.
- 3) acts only on the Sertoli cells.
- 4) promotes protein degradation.
- 5) none of the above.

The Endocrine System Lecture 5

E. Reproductive Hormones

- General concepts
- Reproductive hormones in the male
- Reproductive hormones in the female



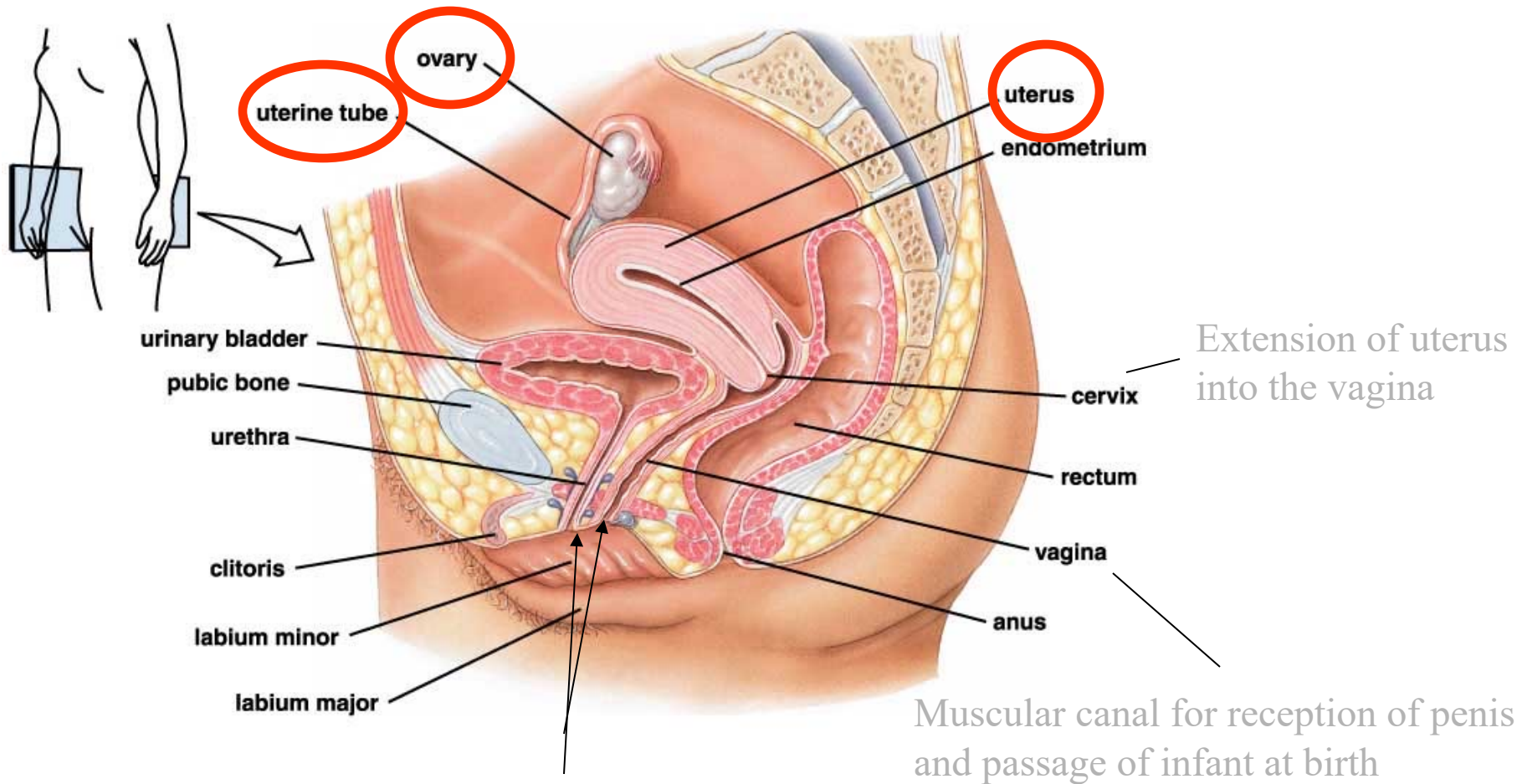
The Sex Organs of the Female

1) *Vulva* (external genitalia)

2) The internal organs include 4 parts:

- Vagina
- The *uterus*, a muscular organ that holds and nourishes the developing fetus during pregnancy.
- Uterine tube (oviduct or fallopian tube): site of fertilization of ovum; transport of ovum from the ovary to uterus.
- Two ovaries produce ova and female sex hormones, primarily estrogens and progesterone.

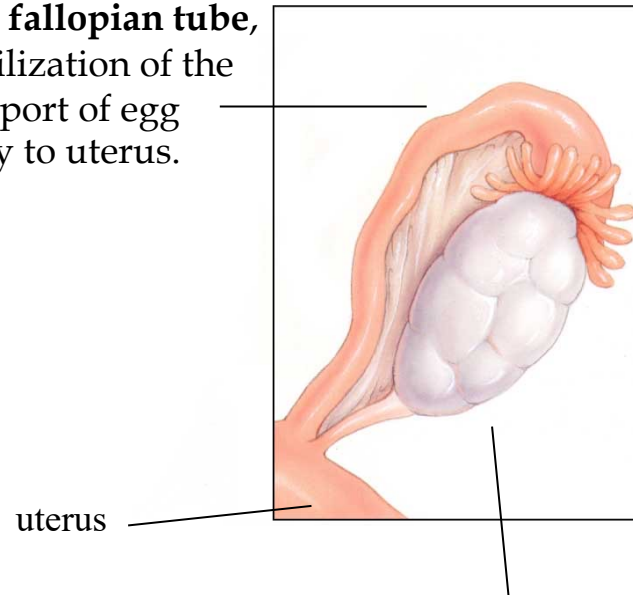
Human Female Reproductive System



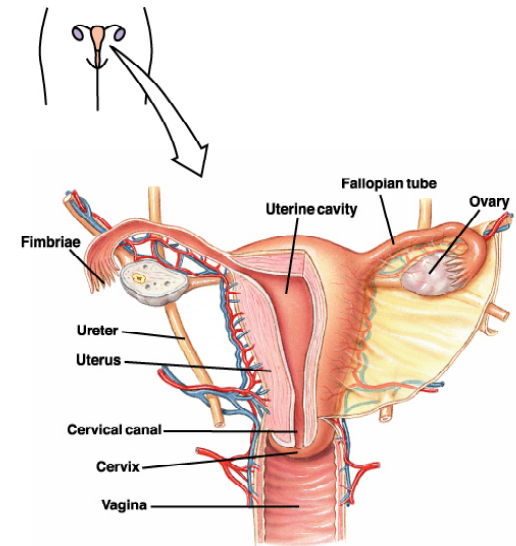
Note separate exits for urinary and reproduction systems

The human female primary sex organ and associated organs

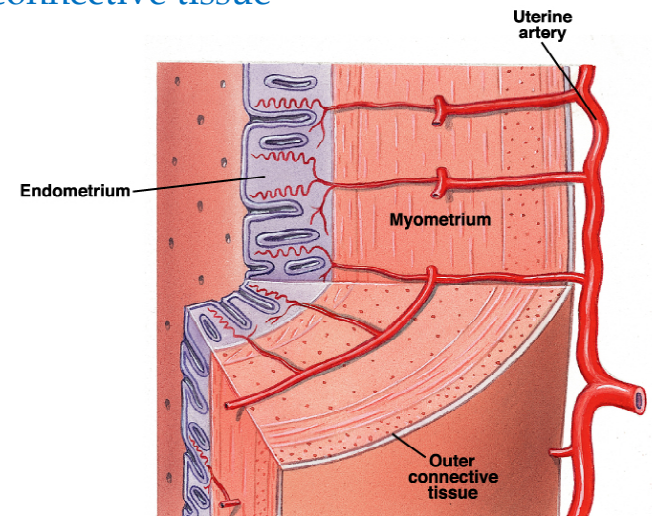
Oviduct = fallopian tube, site of fertilization of the egg; transport of egg from ovary to uterus.



Ovary = primary sex organ; egg production and estrogen and progesterone production all occur within the **follicles**.

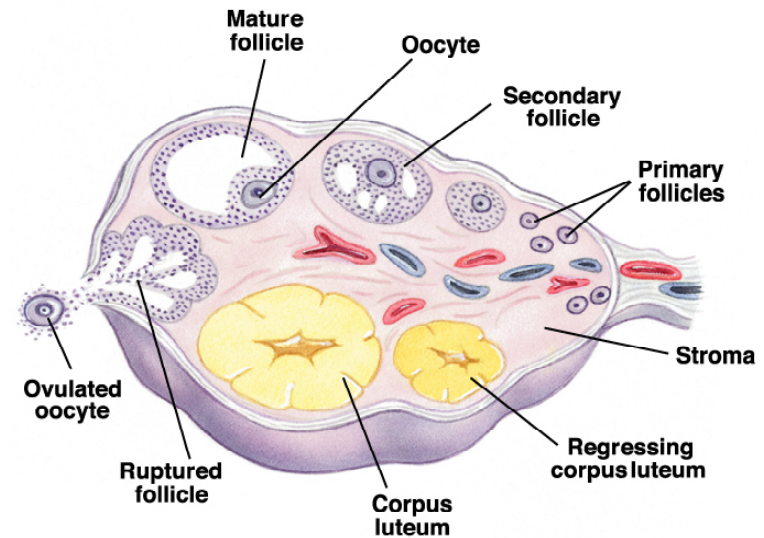


Uterus — Muscular organ for implantation & development of the fetus. The endometrium is a secretory tissue layer made of glands & connective tissue



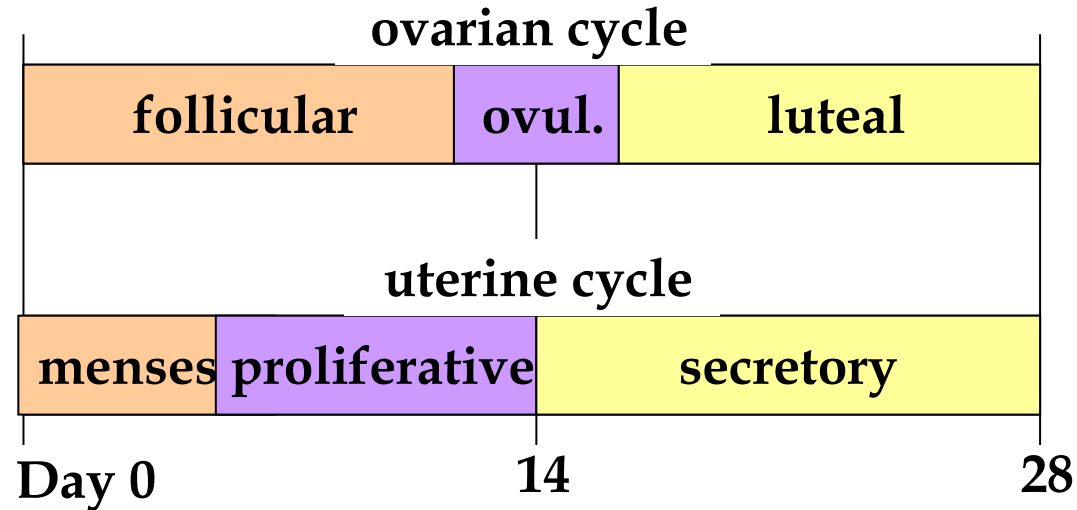
Oogenesis

- The process of the formation, development, and maturation of ova (singular ovum = egg).
- In the middle of each ovulatory cycle (a month), a single ovum is expelled by one of the two ovaries into the uterus during the reproductive life span of the female (puberty to menopause). About 400-500 mature ova will be produced in a woman's reproductive life span.
- They are formed in the primary follicles in the ovary. Follicle = oocyte + supporting cells. The supporting cells provide nutrients, chemical signals, and a barrier.
- After ovulation, the follicle becomes the *corpus luteum*. The corpus luteum will degenerate if no fertilization and implantation of the ovum occur.



Reproductive Cycle

- The purpose is to produce gametes and prepare the uterus for implantation.
- On average 28 days in humans.
- Actually there is two cycles: Ovarian cycle and uterine cycle



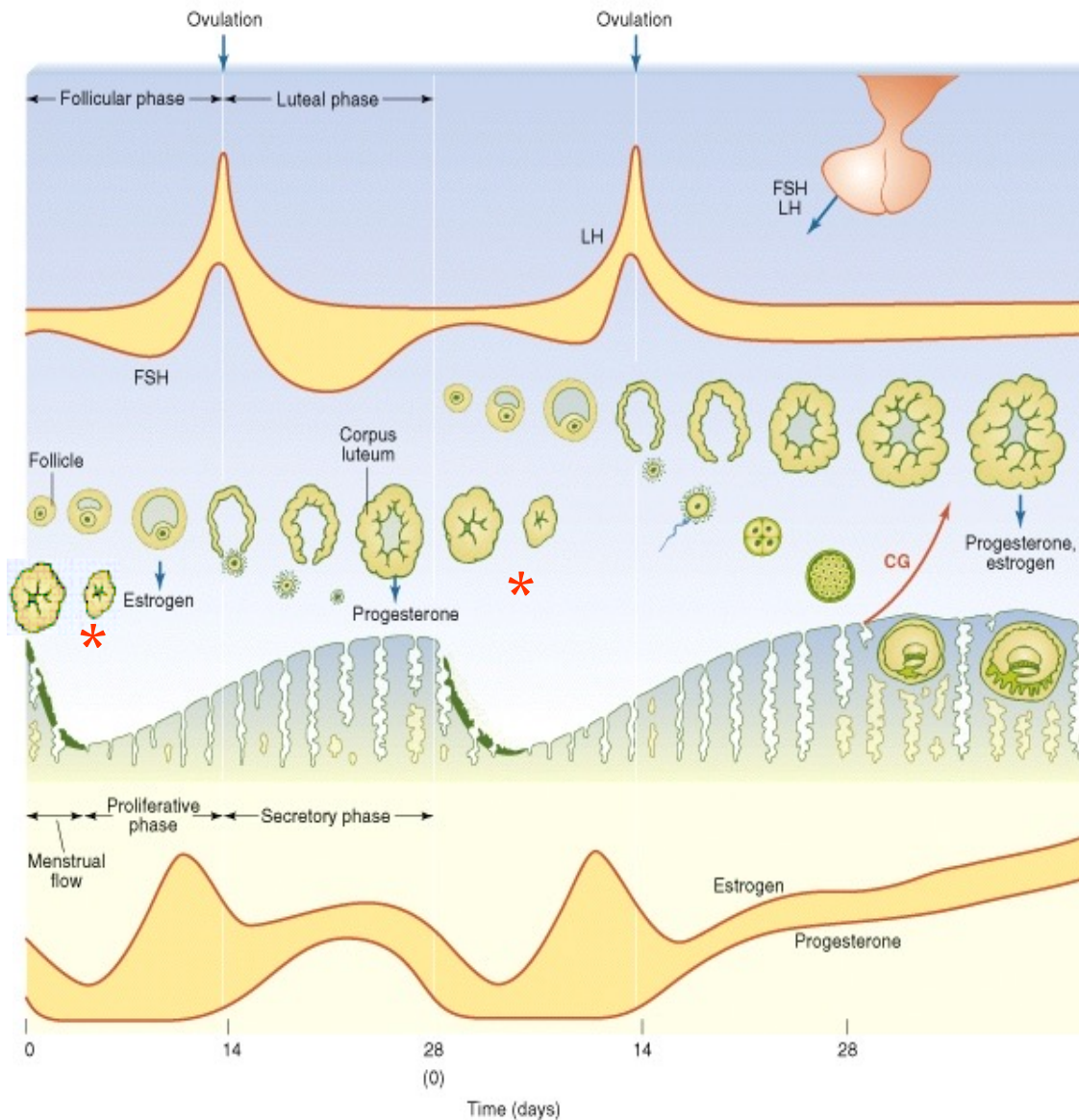
Ovarian Cycle

- Follicular (preovulatory) phase
 - growth & development of follicles in the ovaries
 - lasts 10-14 days.
- Ovulation
 - release of egg around day 14
- Luteal phase
 - transformation of the follicle into the corpus luteum

Uterine Cycle

- menses
 - Menstrual bleeding
- proliferative phase
 - Endometrium adds a new layer of cells (functional endometrium).
- secretory phase
 - corpus luteum hormones enhance endometrium secretions.

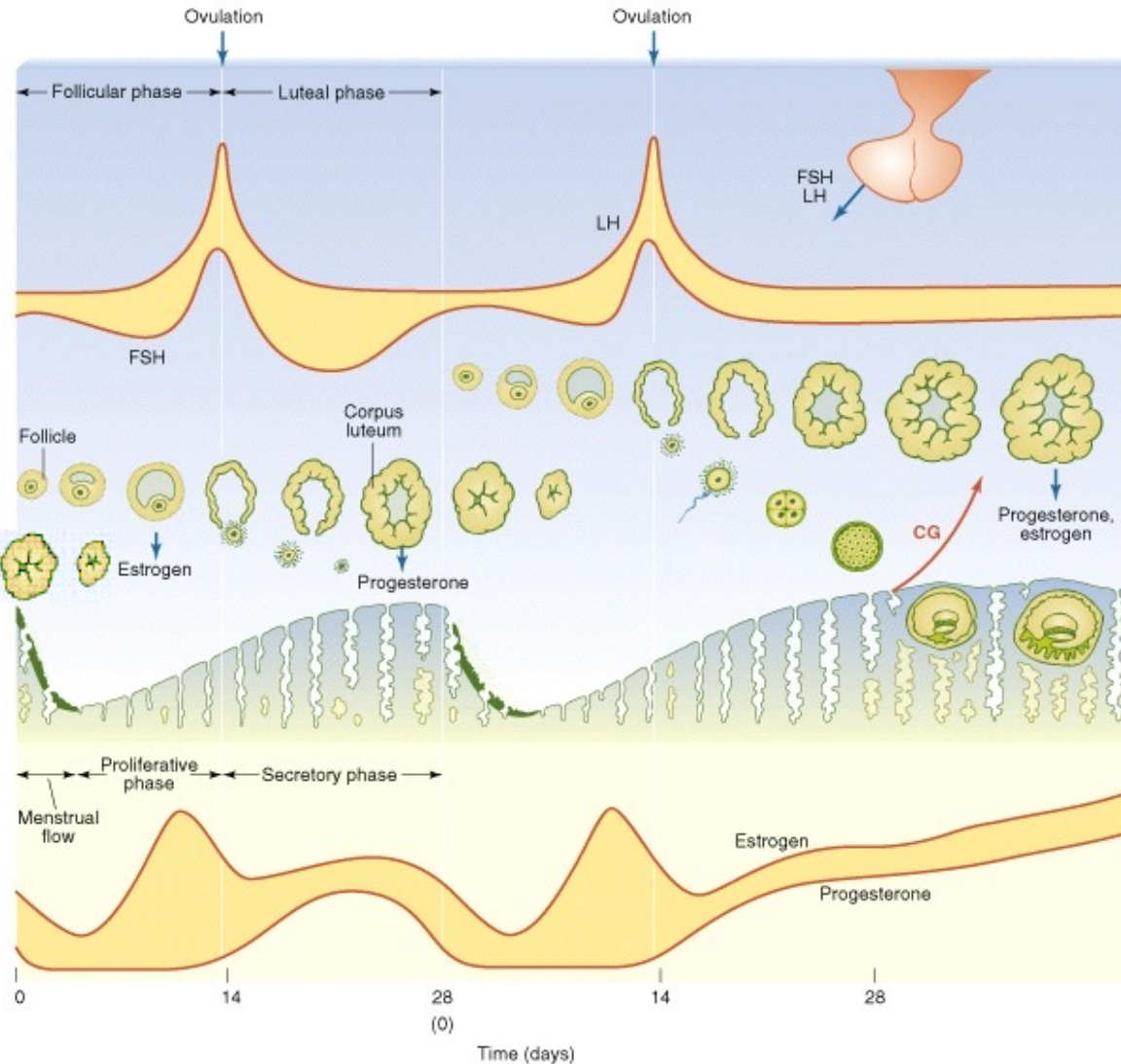
Human reproductive cycle



Day 0-10

- Previous Corpus luteum disintegrating (red asterisks) - no implantation of fertilized ovum, negative feedback on the hypothalamus by hormones secreted from corpus luteum is removed.
- The hypothalamus produces more GnRH, triggers release of FSH and LH.
- Follicle stimulated by FSH, begins to grow and produce estrogen.
- Menstruation- sloughing off of endometrium (lasting ~ 5 days) due to degenerated corpus luteum (estrogen/progesterone fall).

Human reproductive cycle



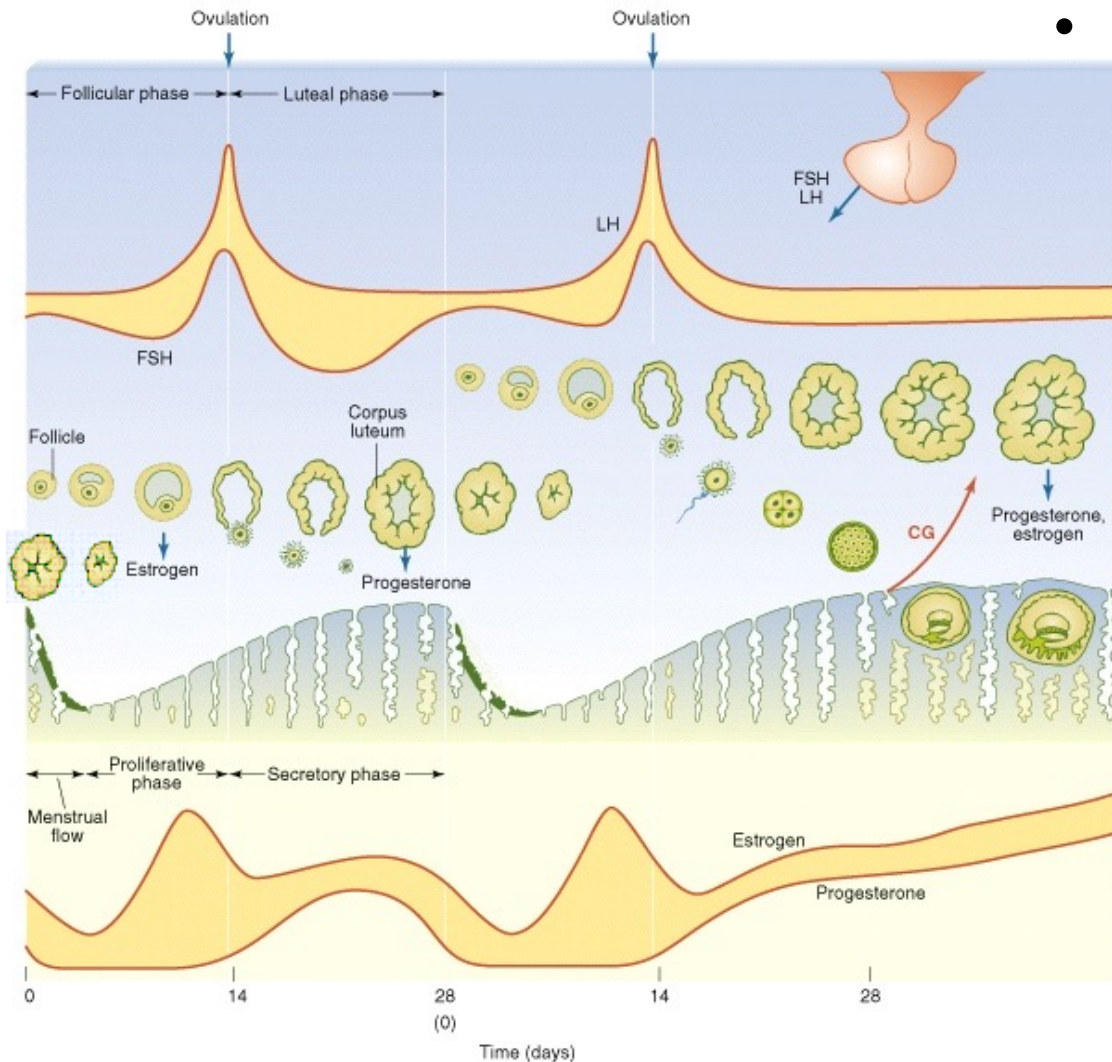
Day 10-12

- Increasing estrogen levels of growing follicle provide **positive** feedback to the hypothalamus.

Day 12 -14

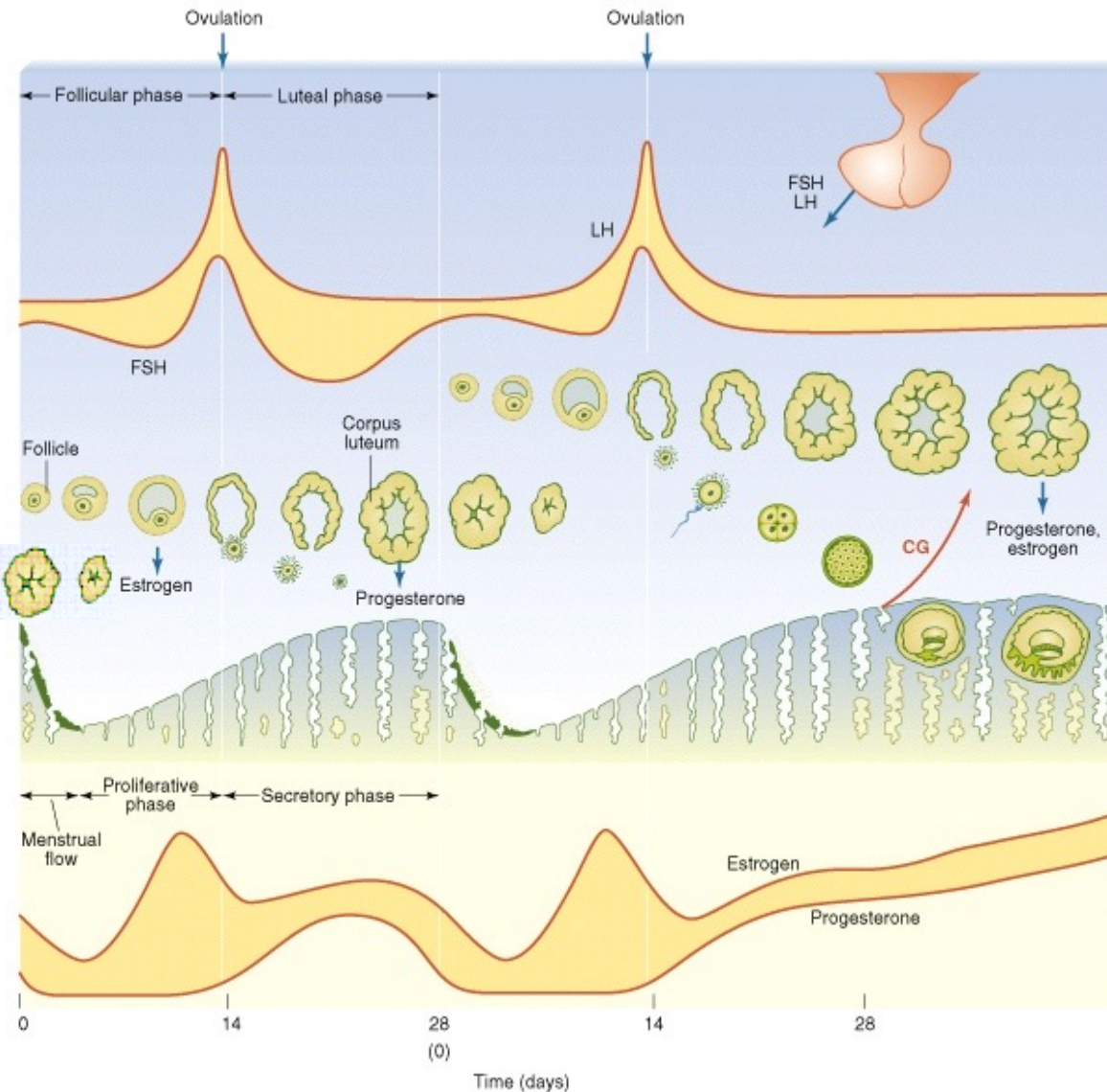
- Increased estrogens lead to buildup of the **endometrium**.
- Increased activation of the hypothalamus leads to a surge of LH and FSH from the anterior pituitary.
- Surge of FSH and LH trigger ovulation.

Human reproductive cycle



- Days 14 - 28 Luteal Phase
 - After release of ovum, follicle develops into corpus luteum.
 - Corpus luteum secretes progesterone and some estrogen.
 - High levels of progesterone/estrogen have **negative** feedback effect on the hypothalamus - decreased secretion of FSH and LH.
 - **Progesterone/estrogen also stimulate buildup of endometrium.**
 - Inhibin, released from follicle at ovulation, also inhibits FSH release.
 - In absence of implantation, corpus luteum degenerates.
 - Progesterone and estrogen levels fall, triggering menstruation again (next cycle).

Human reproductive cycle



- Days 14 - 28 - Pregnancy
 - After release from follicle, ovum travels from the ovary down Fallopian tubes to uterus.
 - Fertilization usually happened in Fallopian tubes, several days after ovulation.
 - If fertilization occurs, the fertilized ovum implants itself in the endometrium.
 - Within a day after implantation, placenta begins to secrete chorionic gonadotropin (CG).
 - CG "rescues" the corpus luteum, preventing it from degenerating.
 - Corpus luteum maintains production of progesterone and estrogen for first trimester, after which placenta secretes these hormones itself.

Female Reproductive Hormones

- Two major hormones:
estrogen and progesterone
- Estrogen includes estradiol, estrone and estriol.

Effects of Estrogen

Growth and Development at Puberty:

- - Stimulates growth of sex organ (1st sex characteristics).
- - Stimulates 2nd sex characteristics.
- - Stimulates growth of long bones and early closure of the epiphyses.

Hormonal

- - Stimulates release of GnRH, LH, & FSH during the follicular phase and **inhibits** release of these same hormones during the luteal phase (see the previous slides).
- - Decreases sensitivity of peripheral tissues to insulin.

Metabolic

- - Protects against bone loss in women.

Gestational Diabetes

- ❑ Due to decreased sensitivity of peripheral tissues to insulin during pregnancy.
- ❑ Usually develops at ~ 20-24 wks. in a pregnancy. As the placenta grows larger, the more estrogen is produced, and the greater the insulin resistance becomes.
- ❑ In most women, the pancreas makes additional insulin to overcome this insulin resistance. When the maximal insulin from the pancreas is still not enough to overcome the effect of the placenta's hormones, gestational diabetes results.

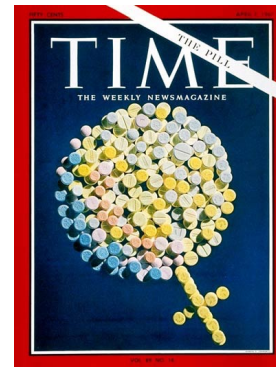


Effects of Progesterone

- Stimulates full development of the endometrium (preparing it for pregnancy).
- Stimulates further development of the mammary glands.
- Provides negative feedback to the hypothalamus to prevent further ovulation, preventing ovulation during pregnancy.
- Antagonizes the effects of testosterone & dihydrotestosterone. Prevents target tissues from responding to androgens in the female.

Contraceptive Pills (“the pill”)

- estrogen + progestogen
- 99.7% effective
- Prevent ovulation by suppressing release of gonadotropins, and inhibit sperm mobility by increasing the viscosity of cervical mucus.
- Very safe, incidences of side effects are noted (the most common is “breakthrough bleeding”).
- The first generation “pill” was approved by FDA in 1960 and the rapid spread of its use generated enormous social impact.



"The pill" is a miraculous tablet that contains as little as one thirty-thousandth of an ounce of chemical. It costs 11¢ to manufacture; a month's supply now sells for \$2.00 retail. It is little more trouble to take on schedule than a daily vitamin. Yet in a mere six years it has changed and liberated the sex and family life of a large and still growing segment of the U.S. population: eventually, it promises to do the same for much of the world.

"The pill," as oral contraceptives are now universally known, may well have...

The Pill | Apr. 7, 1967

Hormones and in vitro fertilization



Patrick Steptoe



Robert Edwards



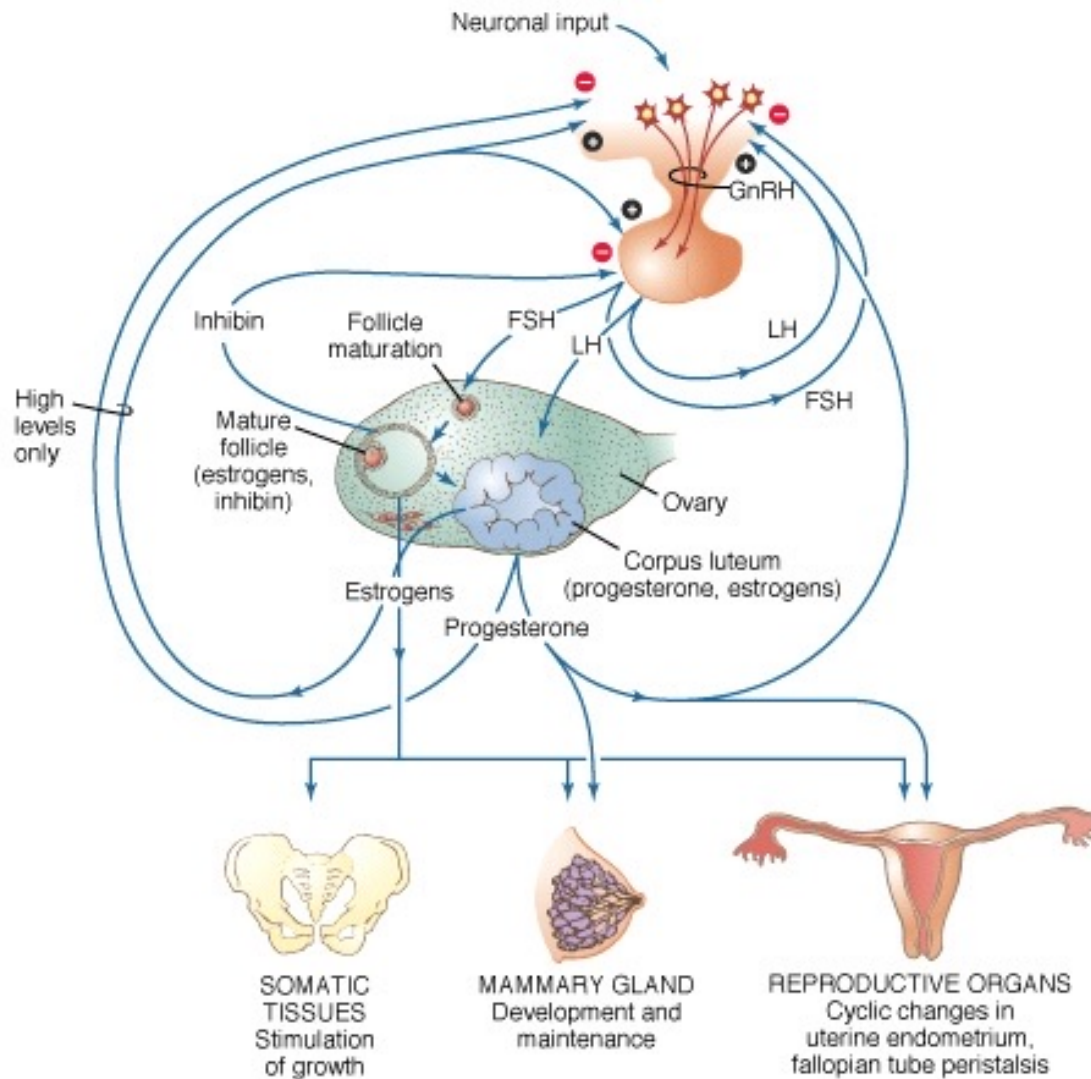
Louise Brown (1978-), the first “test tube baby”

IVF invented by Robert Edwards (Nobel Prize, 2010) and Patrick Steptoe

> 8 million IVF babies born since 1978

IVF routinely needs ovarian hyperstimulation to generate multiple eggs, which needs FSH. And final maturation induction of oocytes needs human chorionic gonadotropin (hCG).

Unwanted embryos are sometimes used in stem cell research, which is ethically controversial.



Reproductive Hormones (Summary)

Quizzes

The function of the oviducts is :

- 1) Secreting estrogen
- 2) The site of implantation
- 3) The site of fertilization
- 4) The site of ova production
- 5) None of the above

Ovulation can be prevented by:

- 1) High level of estrogen and progesterone
- 2) FSH
- 3) LH
- 4) GnRH
- 5) None of the above

Which of the following is FALSE concerning the female reproductive cycle?

- 1) LH secretion is inhibited by estrogen and progesterone in the luteal phase of the cycle
- 2) The corpus luteum secretes estradiol and progesterone
- 3) FSH release is partly responsible for follicular maturation
- 4) The corpus luteum does not regress toward the end of the reproductive cycle if human chorionic gonadotropins are produced
- 5) Ovulation is prevented by follicular estrogen secretion.

Gestational diabetes is caused by:

- 1) ingestion of excessive carbohydrates during pregnancy
- 2) high level of progesterone during pregnancy
- 3) high level of estrogen during pregnancy
- 4) increased glucose absorption in the GI tract
- 5) excessive secretion of glucagon